

Academic integrity in the age of Artificial Intelligence (AI) authoring apps

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Abstract

What does it mean to write, learn to write, and teach writing in an age when students can use the latest artificial intelligence (AI) co-authoring tools to produce entire essays without even adding an original idea or composing a single sentence? This article addresses questions of authorship and academic integrity concerning the use of AI writing assistants and the latest GPT-3 (Generative Pre-trained Transformer, Version 3) tools. It begins by problematizing the use of these tools, and then illustrates how students can use these tools to paraphrase, summarize, extend, and even create original texts with minimal original input, raising questions about authorship and academic integrity. The author argues that as these tools become more widespread, teachers must find creative ways to integrate them into the teaching and learning process and offer practical suggestions for classroom practice. The author hopes to raise awareness about threats to academic integrity brought about by the use of the latest AI co-authoring tools and aims to equip teachers with strategies to embrace the use of these new digital technologies in the teaching of writing.

1 | INTRODUCTION

The launch of OpenAI's (<https://openai.com/api/>) ChatGPT in November 2022 has created a furore in fields as divergent as computer programming, healthcare, human resource management, entertainment, and education. The foundation of ChatGPT is GPT-3 or Generative Pre-trained Transformer, Version 3, a deep learning neural network model which uses artificial intelligence (AI) to train computers to generate human-like texts based on data sets from the internet and other sources (Floridi & Chiriatti, 2020; Zhang & Li, 2021). ChatGPT is the latest and arguably most sophisticated publicly available version of an AI co-authoring tool. Such tools allow authors to compose complete texts with minimal input. Using such *augmented writing* tools (Hellman, 2019) for academic writing, authors simply provide a title or pose a question. These GPTs will suggest sentences that develop the topic relevantly and coherently, leaving the author merely to select sentences until the task is complete. Some tools, such as OpenAI's Text Completion feature, can even write an entire essay in response to a simple prompt.

The widespread availability of such tools is a serious concern in education (Godwin-Jones, 2021) because it raises questions about authorship and academic integrity in assessment. Their use is worrying for education in general, especially as educators seek to equip learners with 21st century competencies and values such as critical and inventive thinking, information literacy skills, resilience, and integrity. Although these tools may take the pain out of the research and writing process, they deprive learners of the opportunity to develop academic writing skills and the cognitive, linguistic, and socioemotional competencies that they could gain through employing and experiencing authentic processes of academic writing. Most of all, students are denied opportunity for self-expression and creativity. Biermann (2022) found that co-writing between humans and AI may threaten the author's control, autonomy, and ownership. Finally, the cost of these tools raises concerns about equity and access. While most offer free versions, the paid premium version unlocks many additional affordances, such as auto-citation and plagiarism checks that can enhance the final written output and prevent detection of cheating, thus advantaging the privileged and potentially widening the educational achievement gap.

The widespread use of GPTs (generative pre-trained transformers) is especially worrying in language teaching and learning. As second or foreign language students often face difficulties with academic writing (Phakiti & Li, 2011; Tang, 2012), they may be tempted to rely on such tools when faced with pressing assignment deadlines. However, to become proficient writers in English, learners need to apply their knowledge of the nature of academic texts, making use of academic vocabulary, grammar, sentence, and rhetorical structures to express their ideas for a specific purpose and audience (Swales & Feak, 2010). Furthermore, they need to make use of the processes of brainstorming, drafting, revising, editing, and rewriting to develop academic vocabulary (Muncie, 2002), paragraph structure (Martínez et al., 2020), and their overall writing abilities (Huang & Zhang, 2020).

For these reasons, the use of AI authoring tools, especially the new generation of co-authoring GPTs, is a pressing issue in TESOL and warrants urgent attention. This is especially so because the reaction of educators to the release and subsequent use of ChatGPT has been largely negative, with some institutions reverting to traditional methods of assessment such as supervised pen and paper exams (Cassidy, 2023a, 2023b; Yang, 2023). Yet the use of such tools can bring about a host of benefits for language learners. For example, learners have opportunities to produce meaning-focused written output to prompt the GPT to generate appropriate texts. In addition, the vast number of texts generated offers an array of meaning-focused, accurate, and authentic language models as well as sources of content knowledge. The aim of this article is to raise awareness about the use of these tools and discuss ways we can use them to enhance learner language

development. To do so, I will describe two types of AI authoring tools for academic writing—a writing assistant, Wordtune, and a GPT, Jenni, to highlight the threat they pose to academic integrity in academic writing and language assessment. I will then suggest ways in which teachers can “befriend the genie,” so to speak, and use these tools as part of writing instruction. Finally, I will identify research areas that can provide educators with a better understanding of how these tools can help or hinder us in teaching writing skills.

2 | FROM WRITING ASSISTANTS TO GENERATIVE PRE-TRAINED TRANSFORMERS (GPTS)

In recent years, the use of intelligent writing assistants has become more widespread (Gayed et al., 2022). The early versions of these tools required authors to input their ideas by constructing or inserting the text, and tools such as Grammarly and Microsoft Editor could offer feedback, which the writer could use to correct or improve the writing. More advanced versions of these tools such as Wordtune (<https://app.wordtune.com/v2/editor/>) and Quillbot (<https://quillbot.com/>) can even revise the text provided by the author. Wordtune, for example, can rewrite the text in casual or formal tones, shorten and expand the length of the text and rewrite entire paragraphs. Serving as intelligent writing assistants, these tools do the work of human copyeditors and editors to improve the writing but, crucially, the ideas still come from the writer.

2.1 | Wordtune

The following describes how authors can use Wordtune to paraphrase and summarize their writing. As shown in Figure 1a, I cut and pasted the abstract of an article from the internet. Next, I highlighted the first sentence and clicked on the “Rewrite” tab. Wordtune offered four choices of paraphrased sentences to select from (Figure 1b). After selecting one, the paraphrased text replaced the original text from the original article (Figure 1c). As Figure 1c shows, Wordtune captured the meaning of the original text accurately. The paid version of this tool allows authors to select the tone (casual or formal), shorten or expand the text, and rewrite whole paragraphs instead of one sentence at a time. There is also no limit to the number of daily rewrites, whereas the free version has a daily limit. (For a review on the affordances and limitations of Wordtune, see Zhao, 2022).

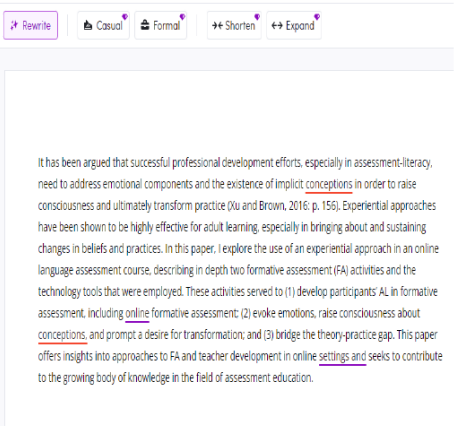
2.2 | Jenni

More recently, a new generation of AI authoring tools has emerged, namely *generative pre-trained transformers* (GPTs). Describing the power of the latest GPT tool, Godwin-Jones (2021) explained, “simplicity belies its power and versatility. GPT-3 can not only complete a phrase or sentence coherently, it can generate connected discourse of considerable length” (p. 6). Hellman (2019) provides a more poetic description about how these “co-writing” tools work:

Like paramours who know each other so intimately they finish one another's sentences, the software takes a simple phrase or sentence typed into the text box and offers to transform it into a rich paragraph. The writer still wields control of the document by choosing whether to implement a suggestion or edit the text.

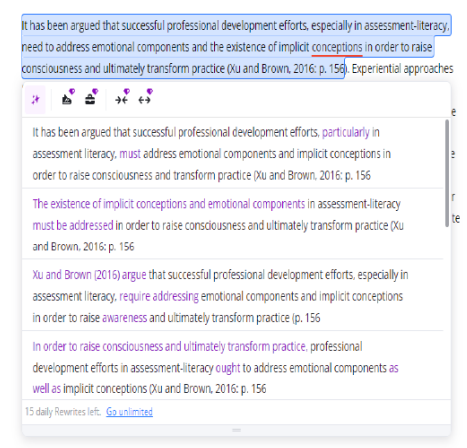
How to use Wordtune to paraphrase text cut and pasted from the internet

a: Original abstract cut and paste from the internet



It has been argued that successful professional development efforts, especially in assessment-literacy, need to address emotional components and the existence of implicit conceptions in order to raise consciousness and ultimately transform practice (Xu and Brown, 2016: p. 156). Experiential approaches have been shown to be highly effective for adult learning, especially in bringing about and sustaining changes in beliefs and practices. In this paper, I explore the use of an experiential approach in an online language assessment course, describing in depth two formative assessment (FA) activities and the technology tools that were employed. These activities served to (1) develop participants' AI in formative assessment, including online formative assessment; (2) evoke emotions, raise consciousness about conceptions, and prompt a desire for transformation; and (3) bridge the theory-practice gap. This paper offers insights into approaches to FA and teacher development in online settings and seeks to contribute to the growing body of knowledge in the field of assessment education.

b: Wordtune offers paraphrase sentence by sentence



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It has been argued that successful professional development efforts, particularly in assessment literacy, must address emotional components and implicit conceptions in order to raise consciousness and transform practice (Xu and Brown, 2016: p. 156)

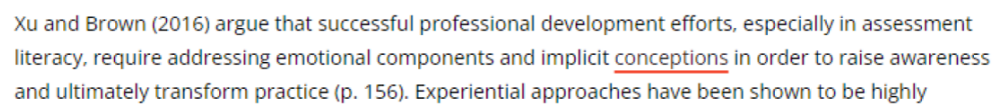
The existence of implicit conceptions and emotional components in assessment-literacy must be addressed in order to raise consciousness and ultimately transform practice (Xu and Brown, 2016: p. 156)

Xu and Brown (2016) argue that successful professional development efforts, especially in assessment literacy, require addressing emotional components and implicit conceptions in order to raise awareness and ultimately transform practice (p. 156)

In order to raise consciousness and ultimately transform practice, professional development efforts in assessment-literacy ought to address emotional components as well as implicit conceptions (Xu and Brown, 2016: p. 156)

15 daily Rewrites left: [Go unlimited](#)

c: Wordtune replaced the sentence with the chosen paraphrase



Xu and Brown (2016) argue that successful professional development efforts, especially in assessment literacy, require addressing emotional components and implicit conceptions in order to raise awareness and ultimately transform practice (p. 156). Experiential approaches have been shown to be highly

FIGURE 1 a: Original abstract cut and paste from the internet. b: Wordtune offers paraphrase sentence by sentence. c: Wordtune replaced the sentence with the chosen paraphrase

An example of this tool is [jenni.ai](#), described on its website as “the ultimate content assistant” that writers can use to “autocomplete” their writing. Unlike writing assistants, these tools only require the writer to provide information about the intended topic, text type, and style (see [Figures 2a](#) and [2b](#)) and the tool will then suggest content by offering sentences that are relevant to and coherently develop the topic. The author simply has to select the next and subsequent sentences until the task is completed (see [Figure 2c](#) and the appendix). To be clear, the authors do not provide their own ideas but are simply choosing from a range of options. Authors can even expand, shorten, paraphrase, or simplify entire paragraphs (see [Figure 2d](#)). Authors can also check that their “original” essay will evade plagiarism detection (see [Figure 2e](#)). There is also a built-in tool to insert citations, but this is in beta version (when I tried it, the references offered were not always credible). The key difference between the free and paid versions is the number of words users are allowed to generate as all functionalities are available in both versions.

How to use Jenni to auto-generate an essay

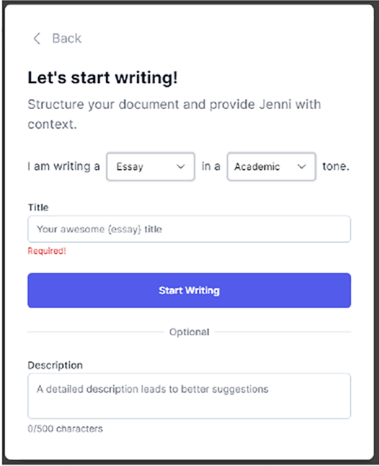
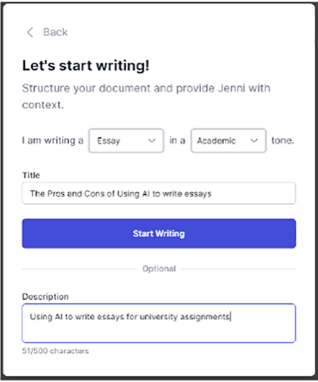
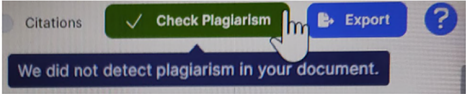
Figure 2a: I selected the text type (essay) and tone (academic) 	Figure 2b: I put in the title and provided a sentence to describe the topic, then clicked on “Start Writing” 
Step 2c: The tool offers a sentence and I can accept the suggestion or seek different suggestions until the essay is completed. The Pros and Cons of Using AI to write essays Many people use artificial intelligence (AI) to write essays for university assignments because it is believed that the technology can help to speed up the process. 	Step 2d: The tool allows me to highlight whole paragraphs and expand, shorten, paraphrase or simplify the paragraph. The Pros and Cons of Using AI 
Step 2e: After completing the essay, I clicked on the “Check Plagiarism” tab and the tool provided a report. 	

FIGURE 2 a: I selected the text type (essay) and tone (academic). b: I put in the title and provided a sentence to describe the topic, then clicked on “Start Writing”, Step 2c: The tool offers a sentence and I can accept the suggestion or seek different suggestions until the essay is completed, Step 2d: The tool allows me to highlight whole paragraphs and expand, shorten, paraphrase or simplify the paragraph, Step 2e: After completing the essay, I clicked on the “Check Plagiarism” tab and the tool provided a report.

3 | ACADEMIC INTEGRITY AND “AUTHORSHIP”

The use of AI writing tools raises questions of academic integrity related to authorship and plagiarism. A recent study by Mellar et al. (2018) highlighted internet copying/pasting, ghost writing, and plagiarism as prevalent authorship issues. How do we define *authorship* and *plagiarism* in an age of AI writing assistants and GPTs? The Collins Dictionary defines authorship as “the origin or originator of a written work, plan, etc.,” while the Cambridge Dictionary defines it as “the state or fact of being the person who wrote a particular book, article, play, etc.” Notably, in neither does the idea of *originality* appear. In academic publishing, authorship pertains to the conception and design of a paper; data collection, analysis, and interpretation; and responsibility for the submission process (Nakazawa et al., 2022). We could argue that writing assistants such as Grammarly and Wordtune simply assist authors to improve the accuracy and style of their writing. The paraphrased words, phrases, and sentences suggested by these tools are based on the text that the author has composed or inserted (possibly by cutting and pasting from other sources, which the author might or might not cite), so, to some extent, the author has “originated” the text and can therefore claim authorship. Would this be considered plagiarism?

Although Husain et al.’s (2017, p. 188) review article on plagiarism found “no consistencies regarding what constitutes plagiarism and how it should be avoided,” Pecorari (2010, cited in Flowerdew & Li, 2007, p. 235) identified six key elements from 53 definitions of plagiarism: “(1) material that has been (2) taken from (3) some source by (4) someone, (5) without acknowledgment and (6) with/without intention to deceive.” Pecorari’s later (2010, p. 4) definition of “textual plagiarism” as “the use of words and/or ideas from another source, without appropriate attribution” highlights the need to reference both words and idea. Swales and Feak (2010, p. 125) define plagiarism as a “deliberate activity” in which students engage in “conscious copying from the work of others.” The word *conscious* suggests that there is intentionality involved. Based on these definitions and in the context of generative AI authoring tools, it could be argued that using ideas and language from another source without proper attribution of both the ideas and language and with or without intent to deceive could be regarded as plagiarism. Therefore, if writers use GPTs which are able to provide auto-citations to reference the ideas, they may not have plagiarized the content. However, they would also have to declare that the sentences were auto-generated by the software. It is unlikely that students would be willing to do this as most teachers would not accept assignments completed in this manner.

The use of AI writing assistants and GPTs raises serious concerns about assessment. The validity, reliability, and fairness of the assessment process will be at risk if teachers assess learners summatively based on assignments that are written either partly or wholly by these tools. Abd-Elal et al.’s (2022) research found that academics with no knowledge of GPTs had difficulty identifying computer-generated writing, which could result in unreliable and unfair scoring of written assessment. The research also found that even by attending one training session, their capacity for detection improved, suggesting the need to provide professional development that raises awareness of possible use of GPTs among students.

3.1 | Using AI authoring tools in teaching and learning

While teachers may have misgivings about the use of writing assistants and GPTs, the use of such tools is likely to become more widespread as a greater number of cheaper and more powerful tools

become available. As it is difficult to prevent students from using these and for teachers to detect computer-generated writing, educators will need to rethink ways of teaching and assessing that are compatible with the use of these technologies (Sharples, 2022, para. 25). While there may be a temptation to throw the proverbial baby out with the bath water by banning technology altogether and returning to more traditional forms of assessment such as supervised pen-and paper tests and oral assessments, this would go against the grain of current beliefs about authentic and alternative forms of assessment. Instead, Sharples (2022) suggests “imaginative” ways to integrate texts generated by AI into teaching: getting learners to evaluate GPT-generated essays against success criteria, then critiquing and revising the writing; using GPTs to co-write, with students writing one paragraph and the GPT the next. These recommendations align with recent moves toward Assessment for Learning (AfL), where the purpose is to improve rather than measure learning by promoting peer interaction, feedback, monitoring, and reflection (Black & Wiliam, 2009).

My own suggestions for using GPTs include the following:

3.1.1 | For teaching

- Getting students to work in pairs or groups to use GPTs to write an essay. This necessitates peers to discuss which sentence best fits the audience, purpose, and text type. Such discussion that occurs during collaborative writing has been shown to increase opportunities for language learning (Wigglesworth & Storch, 2012).
- Getting students to revise specific aspects of language, for example, linking words, passive sentences, and nominalizations, and to attend to targeted aspects of language in an essay generated by a GPT. Such activities promote noticing (Leow, 2019) and language-focused learning (Nation, 2007).
- Getting students to analyze texts on the same topic but generated by the GPTs in different text types and tones (e.g., an essay in an academic tone; a blog or email in a professional, friendly, persuasive or bold tone). By doing this, students can gain an awareness of genres and writing styles.
- Getting students to screen-record the process of creating the texts, so that they can later explain and reflect on their decision to select or change the sentences suggested by the tools. Lindgren and Sullivan's (2003) research found that stimulated recall triggered noticing and language awareness with second language (L2) writers. Looking at their decision-making processes could also promote reflection on decision making, leading to the development of metacognitive strategies of planning, monitoring and evaluating their writing process.

3.1.2 | For assessment

Because teachers may find it difficult to identify the authorship of the final product (Abd-Elaal et al., 2022), prioritizing the process seems more valuable. Therefore, assessment approaches such as portfolio assessment (Curtis, 2017; Lam, 2018) and assessment as learning activities that increase feedback literacy such as those described in Yeo (2021) may shift the emphasis from judging the product to learning from the process. There is a danger here, however, that students will not be able to distinguish the ethicality of using AI-authoring tools at different stages of learning. While it may be acceptable to use them as part of the learning process without concerns about plagiarism, it would not be acceptable to submit auto-generated texts for summative assignments. In addition, if students become reliant on tools to generate written output, they are

deprived of the learning opportunities that come from retrieving language, pushing output, noticing language form, and monitoring output, all necessary for successful language learning. This may erode learners' ability to compose original sentences accurately and fluently.

3.2 | Areas for further research

As AI authoring tools, particularly GPTs, are still relatively new, there are many useful areas for research involving students, teachers, researchers, and even technology developers through research questions such as the following:

Students

- Why and how do students use these tools?
- What are students' perceptions about the ethics of using such tools?
- In what ways and to what extent can these tools improve students' language, reasoning, and critical thinking skills?
- What are the comparative benefits of using different AI writing tools, namely, traditional writing assistants such as Grammarly, early GPTs such as Wordtune, and the most recent models such as jenni.ai or ChatGPT?

Teachers

- What are some ways teachers can use these tools to support language learning?
- How much do teachers know about AI authoring tools and their affordances?
- How can teachers detect computer-generated texts?
- What is the impact of training on teachers' ability to integrate the use of such tools?

Researchers

- What methods can be used to investigate students' decision making when using these tools?
- How can the different types of AI authoring tools be classified?

Technology developers

- How can the tools be designed for learning (e.g., keeping track of learners' decisions so that learners can develop metacognition by reflecting on their decisions)?
- How can the auto-citations be improved to select and cite credible sources?

While there may be a tendency to adopt experimental approaches involving control and experimental groups to demonstrate the effects of technology, as research in this field is still in its infancy and there are ideological and ethical issues surrounding generative AI it may be fruitful to carry out in-depth qualitative as well as ethnographic studies alongside controlled experimental studies. Chapelle's (1997) call for educational technology researchers to adopt research methods from cognitive psychology, constructivism, psycholinguistics, and discourse analyses, as well as second language acquisition, is equally relevant to research on AI generative tools. In terms of data sets, technology itself affords us screen-recording tools to capture the way we actually use

the tools rather than relying on self-reported or stimulated-recall data. Therefore, in addition to qualitative data from questionnaires and interviews, observational data from screen recordings or video recordings of students using the tools and corpora of texts generated by learners could provide more reliable sources of data.

4 | LIMITATIONS OF THE STUDY

It is beyond the scope and intention of this article to review the different writing assistants and GPTs on the market. As new and more powerful ones appear all the time, as evidenced by the release of ChatGPT and, even more recently, GPTZero (<https://gptzero.me/>) as a counter-measure, there would be little point in doing this. Furthermore, app developers are constantly improving their tools, so even the affordances of Wordtune and jenni.ai described above may have been eclipsed. Readers can search the terms *AI writing assistants*, *augmented writing*, and *GPTs* to find information about the latest AI authoring tools. A useful description of some current tools can also be found at <https://renaissancerachel.com/best-ai-writing-tools/>.

5 | CONCLUSION

While educators may be concerned about students' use of the latest AI co-authoring tools, it will be difficult, even impossible, to put the genie back in the bottle. AI writing assistants can offer varying degrees of help from correcting linguistic and stylistic errors to paraphrasing, summarizing, and extending text inserted by the writer to generating whole essays based on a short prompt suggested by the author, who then simply chooses sentences to auto-complete the task. The latest iterations such as ChatGPT can even write the entire essay in response to a simple prompt. This raises questions about authorship and academic integrity, concepts that have become increasingly contestable in the age of AI authoring tools. Sutherland-Smith (2016, p. 575) argues,

It is important to revisit our beliefs about what makes up authorial rights because digital technologies contest the very core of what it means to have authorship rights over text. Authorship and originality also underpin the birth, and continued life, of plagiarism in policy and practice.

Notwithstanding concerns about authorship and academic integrity, the use of these tools especially in second and foreign language teaching prompt us to question future definitions of what it means to write, to learn to write, and to teach writing in a digital world. In the future, will we conceptualize writing as the expression of the author's original ideas in textual form or will it be the selection and combination of pre-generated language into a text? Will learning to write entail learning how to use vocabulary, grammar, and sentence and rhetorical structures to express ideas for a specific purpose and audience or how to critically evaluate, select, and justify ideas suggested by an authoring tool? Finally, will the teaching of writing continue to focus on developing writing subskills to enable students to produce essays or communicating creatively through multimodal representation? To answer these questions, teachers need to be aware of ways in which the latest educational technology tools can help or hinder us and embrace the learning possibilities they bring. Instead of trying to put the genie back in the bottle by ignoring their existence or prohibiting students from using such tools, we

should accept and befriend the genie by showing learners how to use AI authoring ethically and gainfully to achieve their learning intentions and goals.

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REFERENCES

- Abd-Elaal, E.-S., Gamage, S. H. P. W., & Mills, J. E. (2022). Assisting academics to identify computer generated writing. *European Journal of Engineering Education*, 47, 725–745. <https://doi.org/10.1080/03043797.2022.2046709>
- Biermann, O. C. (2022). *Writers want AI collaborators to respect their personal values and writing strategies: A human-centered perspective on AI co-writing* (doctoral dissertation). University of British Columbia, Vancouver, Canada. [10.14288/1.0420422](https://doi.org/10.14288/1.0420422)
- Black, P., & Wiliam, D. (2009). Developing the theory of formative assessment. *Educational Assessment, Evaluation and Accountability*, 21, 5–31.
- Cassidy, C. (2023a, January 10). Australian schools “flying blind” on use of ChatGPT and other learning technology. *The Guardian* <https://amp.theguardian.com/australia-news/2023/jan/10/australia-lagging-behind-on-ai-learning-tech-for-classrooms-report-suggests>
- Cassidy, C. (2023b, January 10). Australian universities to return to “pen and paper” exams after students caught using AI to write essays. *The Guardian* <https://www.theguardian.com/australia-news/2023/jan/10/universities-to-return-to-pen-and-paper-exams-after-students-caught-using-ai-to-write-essays>
- Chapelle, C. A. (1997). CALL in the year 2000: Still in search of research paradigms? *Language Learning*, 1(1), 19–43. <https://dx.doi.org/10.1255/25002>
- Curtis, A. (2017). Portfolios. In J. I. Lontas (Ed.), *The TESOL encyclopedia of English language teaching* (pp. 1–7). Hoboken, NJ: John Wiley. <https://doi.org/10.1002/9781118784235.eelt0326>
- Floridi, L., & Chiriatti, M. (2020). GPT-3: Its nature, scope, limits, and consequences. *Minds and Machines*, 30, 681–694. <https://doi.org/10.1007/s11023-020-09548-1>
- Flowerdew, J., & Li, Y. (2007). Plagiarism and second language writing in an electronic age. *Annual Review of Applied Linguistics*, 27, 161–183. <https://doi.org/10.1017/S0267190508070086>
- Gayed, J. M., Carlon, M. K. J., Oriola, A. M., & Cross, J. S. (2022). Exploring an AI-based writing assistant's impact on English language learners. *Computers and Education: Artificial Intelligence*, 3, 100055. <https://doi.org/10.1016/j.caeai.2022.100055>
- Godwin-Jones, R. (2021). Big data and language learning: Opportunities and challenges. *Language Learning*, 25(1), 16. <http://hdl.handle.net/10125/44747>
- Hellman, M. (2019, April 30). Augmented writing technology: A writer's friend or foe? *The Seattle Times* <https://www.seattletimes.com/business/technology/augmented-writing-technology-a-writers-friend-or-foe/>
- Huang, Y., & Zhang, L. (2020). Does a process-genre approach help improve students' argumentative writing in English as a foreign language? Findings from an intervention study. *Reading and Writing Quarterly*, 36, 339–364. <https://doi.org/10.1080/10573569.2019.1649223>

- Husain, F. M., Al-Shaibani, G. K. S., & Mahfoodh, O. H. A. (2017). Perceptions of and attitudes toward plagiarism and factors contributing to plagiarism: A review of studies. *Journal of Academic Ethics*, 15(2), 167–195. <https://doi.org/10.1007/s10805-017-9274-1>
- Lam, R. (2018). *Portfolio assessment for the teaching and learning of writing*. Cham, Switzerland: Springer. <https://doi.org/10.1007/978-981-13-1174-1>
- Leow, R. P. (2019). Noticing hypothesis. In J. I. Lontas (Ed.), *The TESOL encyclopedia of English language teaching* (pp. 1–7). Hoboken, NJ: John Wiley. <https://doi.org/10.1002/9781118784235.eelt0086.pub2>
- Lindgren, E., & Sullivan, K. P. H. (2003). Stimulated recall as a trigger for increasing noticing and language awareness in the L2 writing classroom: A case study of two young female writers. *Language Awareness*, 12(3–4), 172–186. <https://doi.org/10.1080/09658410308667075>
- Mellar, H., Peytcheva-Forsyth, R., Kocdar, S., et al. (2018). Addressing cheating in e-assessment using student authentication and authorship checking systems: teachers' perspectives. *International Journal for Educational Integrity*, 14, 2. <https://doi.org/10.1007/s40979-018-0025-x>
- Martínez, J., López-Díaz, A., & Pérez, E. (2020). Escritura como proceso en la enseñanza de inglés como lengua extranjera. *Revista Caribeña de Investigación Educativa*, 4(1), 49–61. [10.32541/recie.2020.v4i1.pp49-61](https://doi.org/10.32541/recie.2020.v4i1.pp49-61)
- Muncie, J. (2002). Process writing and vocabulary development: Comparing lexical frequency profiles across drafts. *System*, 30, 225–235. [https://doi.org/10.1016/S0346-251X\(02\)00006-4](https://doi.org/10.1016/S0346-251X(02)00006-4)
- Nakazawa, E., Udagawa, M., & Akabayashi, A. (2022). Does the use of AI to create academic research papers undermine researcher originality? *AI*, 3, 702–706. <https://doi.org/10.3390/ai3030040>
- Nation, P. (2007). The four strands. *Innovation in Language Learning and Teaching*, 1(1), 2–13. <https://doi.org/10.2167/illt039.0>
- Pecorari, D. (2010). *Academic writing and plagiarism: A linguistic analysis*. London, UK: Bloomsbury Publishing Plc.
- Phakiti, A., & Li, L. (2011). General academic difficulties and reading and writing difficulties among Asian ESL postgraduate students in TESOL at an Australian university. *RELC Journal*, 42(3), 227–264. <https://doi.org/10.1177/0033688211421417>
- Sharples, M. (2022). Automated essay writing: An AIED opinion. *International Journal of Artificial Intelligence in Education* <https://doi.org/10.1007/s40593-022-00300-7>
- Sutherland-Smith, W. (2016). Authorship, ownership, and plagiarism in the digital age. In T. Bretag (Ed.), *Handbook of academic integrity* (pp. 575–589). Cham, Switzerland: Springer. https://doi.org/10.1007/978-981-287-098-8_14
- Swales, J., & Feak, C. B. (2010). *Academic writing for graduate students: Essential tasks and skills* (2nd ed.). Ann Arbor: University of Michigan Press.
- Tang, R. (2012). *Academic writing in a second or foreign language: Issues and challenges facing ESL/EFL academic writers in higher education contexts*. London, England: Bloomsbury.
- Wigglesworth, G., & Storch, N. (2012). What role for collaboration in writing and writing feedback. *Journal of Second Language Writing*, 21, 364–374. <https://doi.org/10.1016/j.jslw.2012.09.005>
- Yang, M. (2023, January 6). New York City schools ban AI chatbot that writes essays and answers prompts. *The Guardian* <https://www.theguardian.com/us-news/2023/jan/06/new-york-city-schools-ban-ai-chatbot-chatgpt>
- Yeo, M. (2021). “Experiencing theory first-hand was delightful and informative”: Bridging the theory-practice gap in Online Language Assessment Training. *Iranian Journal of Language Teaching Research*, 9(3 (Special Issue)), 93–116. <https://doi.org/10.30466/ijltr.2021.121078>
- Zhang, M., & Li, J. (2021). A commentary of GPT-3 in MIT Technology Review 2021. *Fundamental Research*, 1, 831–833. <https://doi.org/10.1016/j.fmre.2021.11.011>
- Zhao, X. (2022). Leveraging artificial intelligence (AI) technology for English writing: Introducing Wordtune as a digital writing assistant for EFL writers. *RELC Journal*, 00336882221094089.

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Appendix A

Essay generated by jenni.ai¹

A | THE PROS AND CONS OF USING AI TO WRITE ESSAYS

Many people use artificial intelligence (AI) to write essays for university assignments because it is believed that the technology can help to speed up the process. It is also believed that using AI could result in a reduction in errors. However, there are also a number of cons that must be considered before making a final decision about whether or not to use AI software.

A.2 | WHAT ARE THE BENEFITS OF USING AI TO WRITE ESSAYS?

There are certainly many advantages of using AI to write essays. For starters, the essays written by AI are often better structured and more academically sound than those written by humans. This is because AI systems are capable of generating content that is written in the correct format and meets all of the required academic standards. This can save a lot of time and effort on the part of the student, as they do not need to spend time rewriting the essays in order to make sure that they meet the required standards. Another benefit of using AI is that it can be used to proofread the essay before submission. This means that it will be much easier to identify any mistakes and ensure that they do not make it into the final draft. Finally, using AI is much faster than writing the essay by hand. This is something that many students appreciate because it saves them a great deal of time.

A.3 | WHAT ARE THE DRAWBACKS OF USING AI TO WRITE ESSAYS?

While there are a number of benefits that can be gained from using AI to write essays, there are also a number of drawbacks that must also be considered. One of the biggest disadvantages is the possibility that the program could make mistakes when producing the work. This is a problem that can arise when it is not set up correctly, as it can lead to sentences not being completed properly or the wrong information being included within the text. Another problem that can arise when using AI is that the program may be biased based on the user's gender or ethnicity, which could have an affect on the work produced. Thirdly, there is often a disparity between the quality of the work produced by humans and by machines, which means that the quality of the essay can be inconsistent depending on the person or program that created it. Finally, there is also the issue of cost that needs to be taken into consideration.

Although there are many benefits to using AI, there are some key drawbacks which must be taken into consideration before making the decision to use this technology. The main benefit is that it can help you to produce a high-quality essay in a very short space of time, but it can also make mistakes in the process and often results in you receiving a lower grade than you may have if you wrote it yourself.

¹Only the three underlined words were added by the author. The rest of the essay was auto-generated.